MICHIGAN ENGINEERING
COMPUTER SCIENCE AND ENGINEERING
UNIVERSITY OF MICHIGAN

Open Ad-hoc Categorization with Contextualized Feature Learning

Zilin Wang

Prelim Exam Presentation

Advisor: Professor Stella X. Yu

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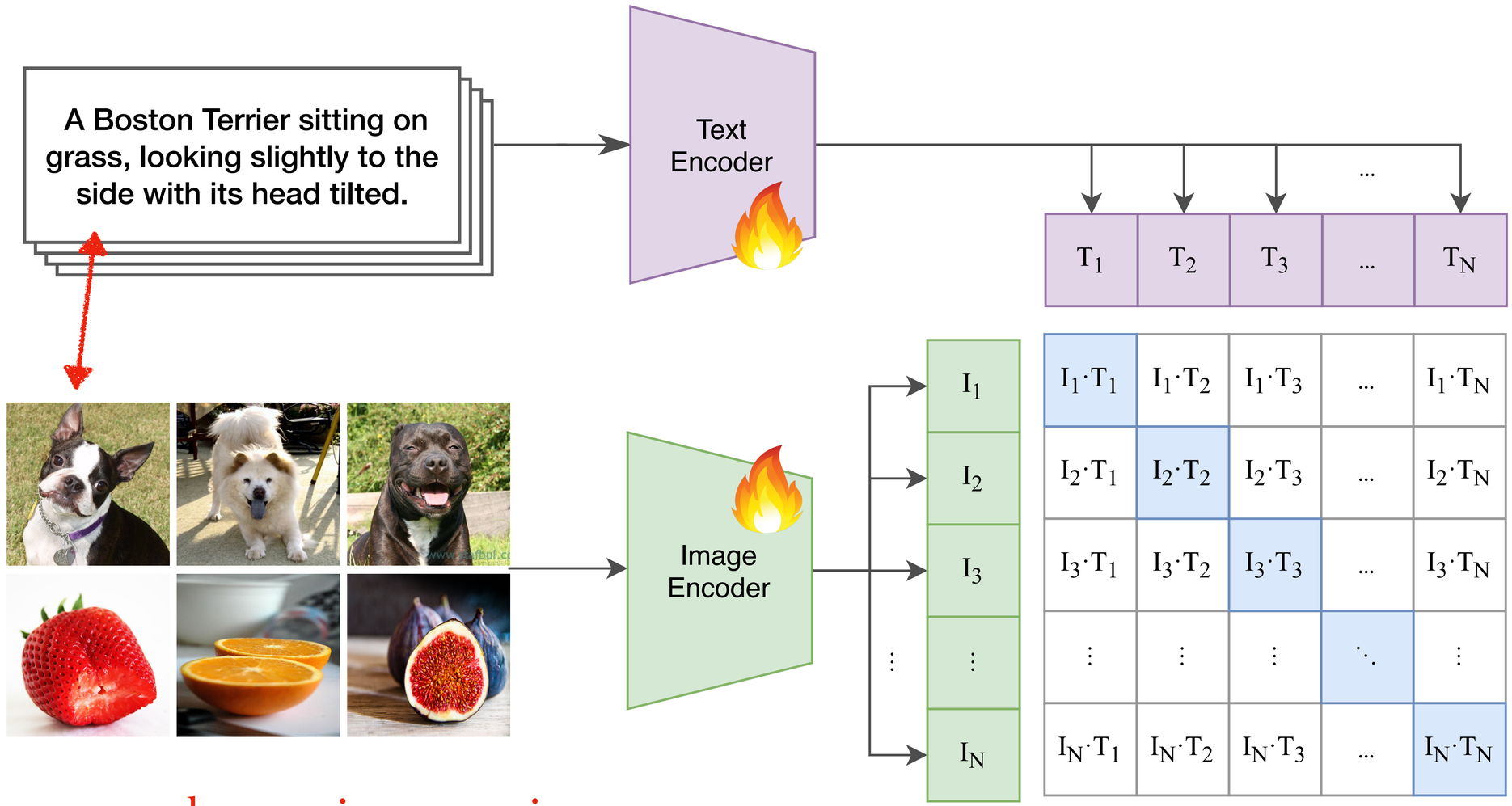
Categorization with Foundation Models

CLIP - Contrastive Language-Image Pretraining



internet-scale text-image pairs

Radford et al,. Learning Transferable Visual Models From Natural Language Supervision, Arxiv 2021

Categorization with Foundation Models

A Boston Terrier sitting on grass, looking slightly to the side with its head tilted.

Radford et al,. Learning Transferable Visual Models From Natural Language Supervision, Arxiv 2021

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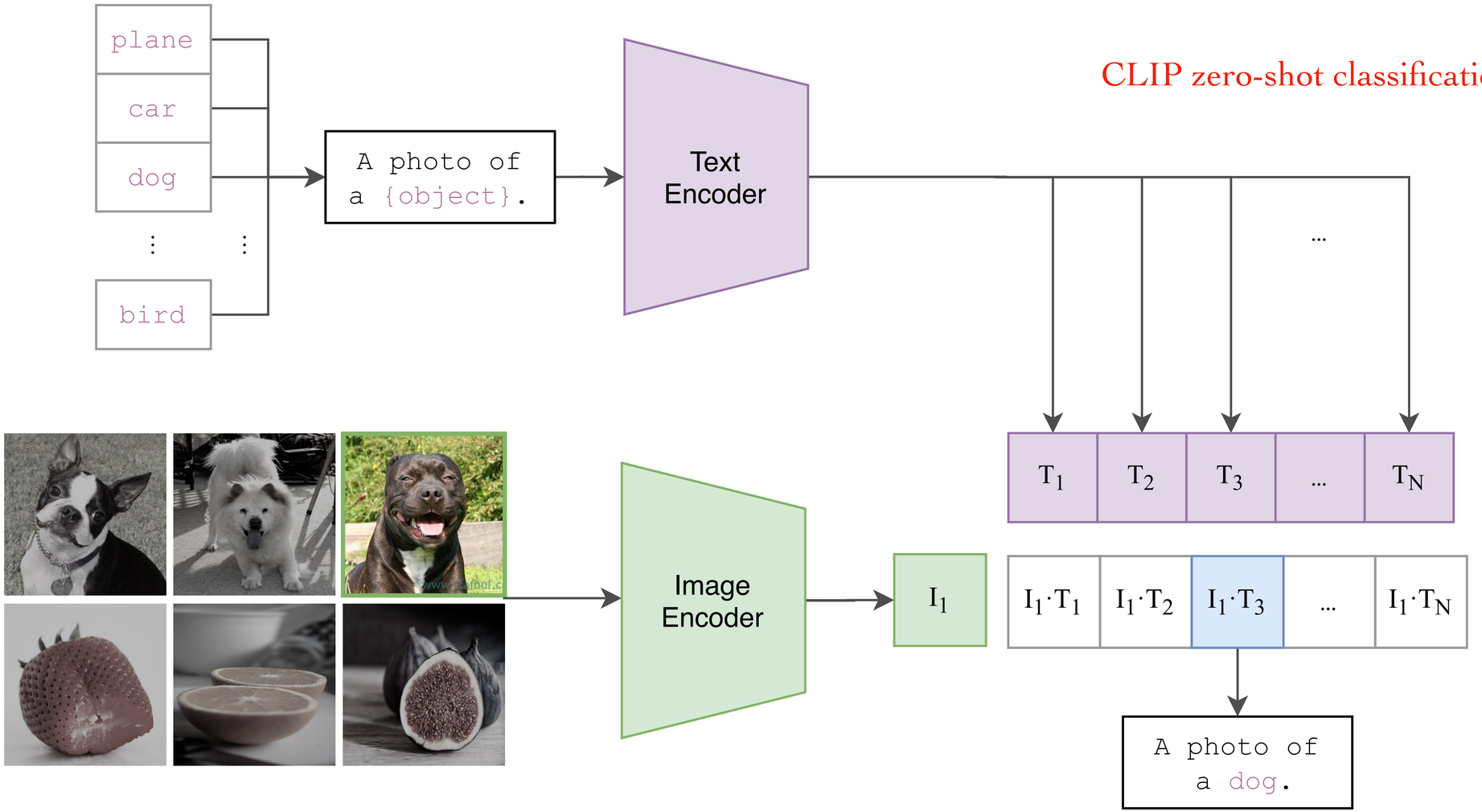
Categorization with Foundation Models



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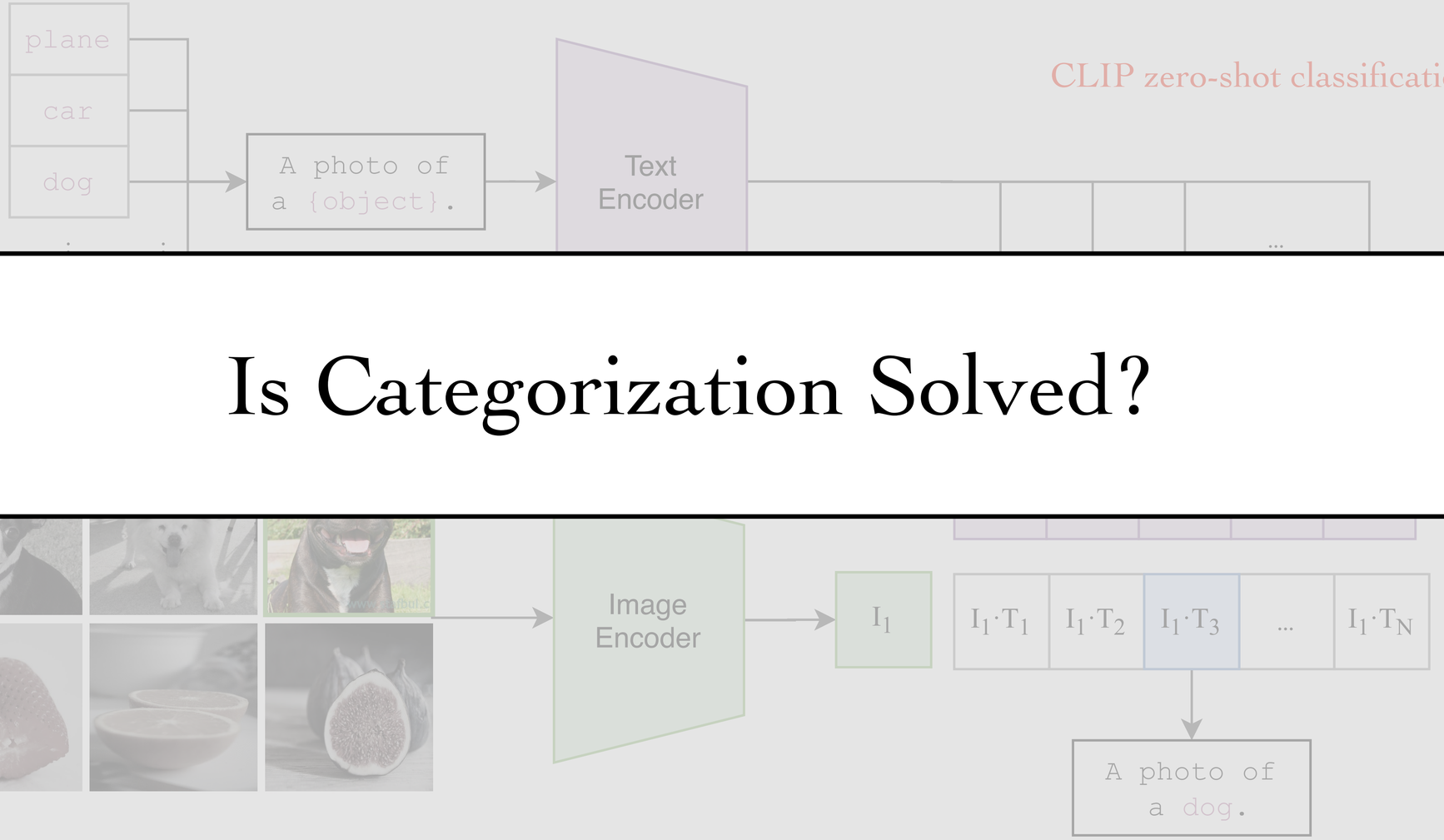
Categorization with Foundation Models



Is Categorization Solved?

CLIP Works Well on **Basic Categories**

but..

(1) Struggles on **Ad-hoc Categories**

- May not share visual and semantic similarities
- Less established in memory and require explicit naming of exemplars
- Often grouped to fit a purpose, dependent on the context
- Recognition relies on the same perceptual processes, but contextualized features

CLIP Works Well on **Basic Categories**

but..

(1) Struggles on **Ad-hoc Categories**

(2) Cannot Generalize to the **Open World**

What are other “Garage Sale Items”?

Open Ad-hoc Categorization (OAK)

Vaze et al,. Generalized Category Discovery, CVPR 2022
Radford et al,. Learning Transferable Visual Models From Natural Language Supervision, Arxiv 2021

OAK Benchmarks

Labeled images
Unlabeled images from **known** classes
Unlabeled images from **novel** classes

Kwon et al,. Image clustering conditioned on text criteria, ICLR 2024
Yao et al,. Human action recognition by learning bases of action attributes and parts, ICCV 2011

OAK Benchmarks

a) Stanford	Action	Location	Mood
Examples	`drinking`, `phoning`	`market`, `residential`	`focused`, `relaxed`
$\lvert\mathcal{Y}_{\mathcal{L}}\rvert$	20	5	2
$\lvert\mathcal{Y}_{\mathcal{N}}\rvert$	20	5	2
$\lvert\mathcal{D}_{\mathcal{L}}\rvert$	320	80	32
$\lvert\mathcal{D}_{\mathcal{U}}\rvert$	9.2K	920	968

b) Clevr-4	Texture	Color	Shape	Count
Examples	`metal`, `rubber`	`red`, `blue`	`torus`, `cube`	`1`, `2`
$\lvert\mathcal{Y}_{\mathcal{L}}\rvert$	5	5	5	5
$\lvert\mathcal{Y}_{\mathcal{N}}\rvert$	5	5	5	5
$\lvert\mathcal{D}_{\mathcal{L}}\rvert$	80	80	80	80
$\lvert\mathcal{D}_{\mathcal{U}}\rvert$	8.3K	8.3K	8.3K	8.3K

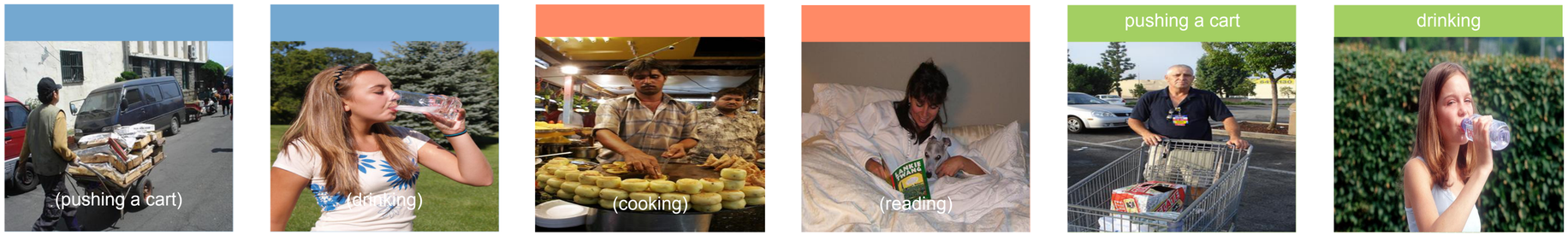
# of known classes = # of novel classes

We use 16 examples for each known class

Vaze et al,. No representation rules them all in category discovery, NeurIPS 2023
Kwon et al,. Image clustering conditioned on text criteria, ICLR 2024
Yao et al,. Human action recognition by learning bases of action attributes and parts, ICCV 2011

CLIP for OAK

Unlabeled known | Unlabeled novel | Labeled



1) Closed-world CLIP

A photo of a person {"*pushing a cart*", "*drinking*"}

2) CLIP + LLM vocabulary

A photo of a person {"*pushing a cart*", "*drinking*", "*carrying a box*", "*eating*"}

ChatGPT

3) CLIP + GT vocabulary

A photo of a person {"*pushing a cart*", "*drinking*", "*cooking*", "*reading*"}

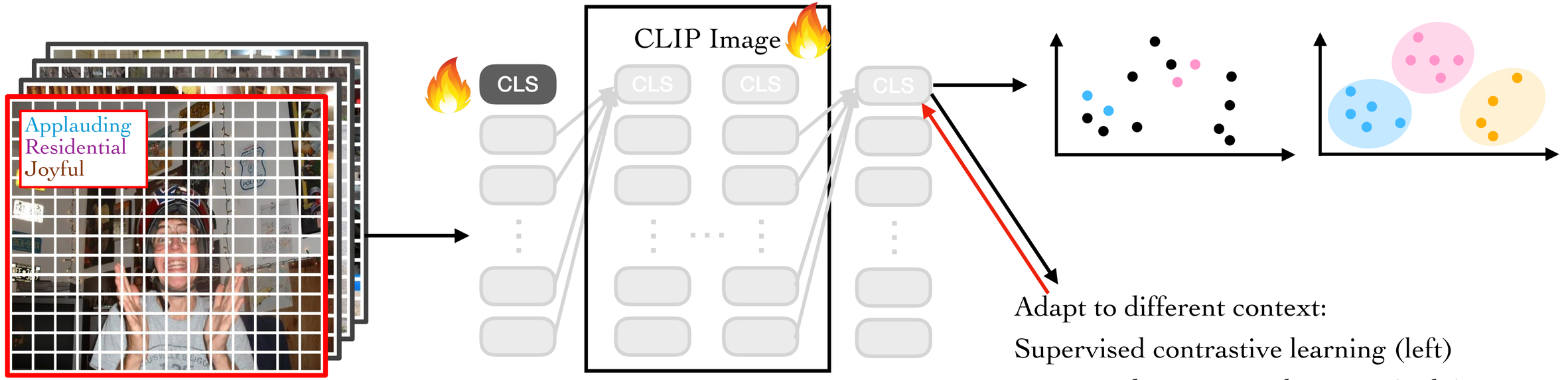
Unrealistic but provides upper bound

CLIP requires novel class names to be known to discover

Radford et al,. Learning Transferable Visual Models From Natural Language Supervision, Arxiv 2021

Generalized Category Discovery (GCD)

1) Extract features using CLIP

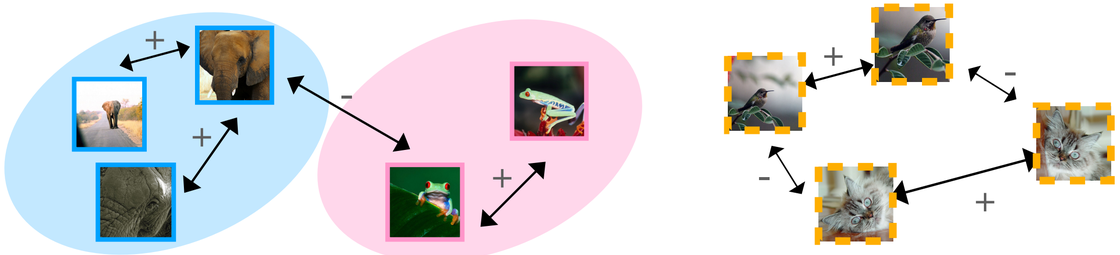


2) Clustering

Adapt to different context:

Supervised contrastive learning (left)

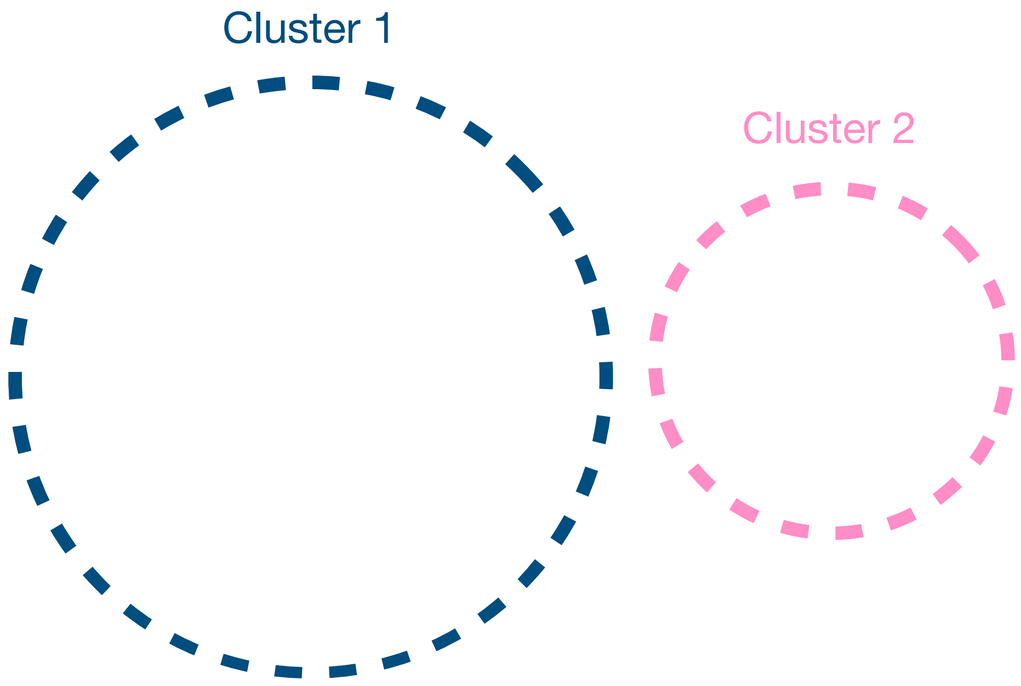
supervised contrastive learning (right)



GCD does NOT require novel class names to discover

Generalized Category Discovery (GCD)

Evaluation through **Hungarian Matching**



	Cluster 1	Cluster 2
Dog	90	10
Cat	80	10

Cluster 1 → Dog

Cluster 2 → Cat

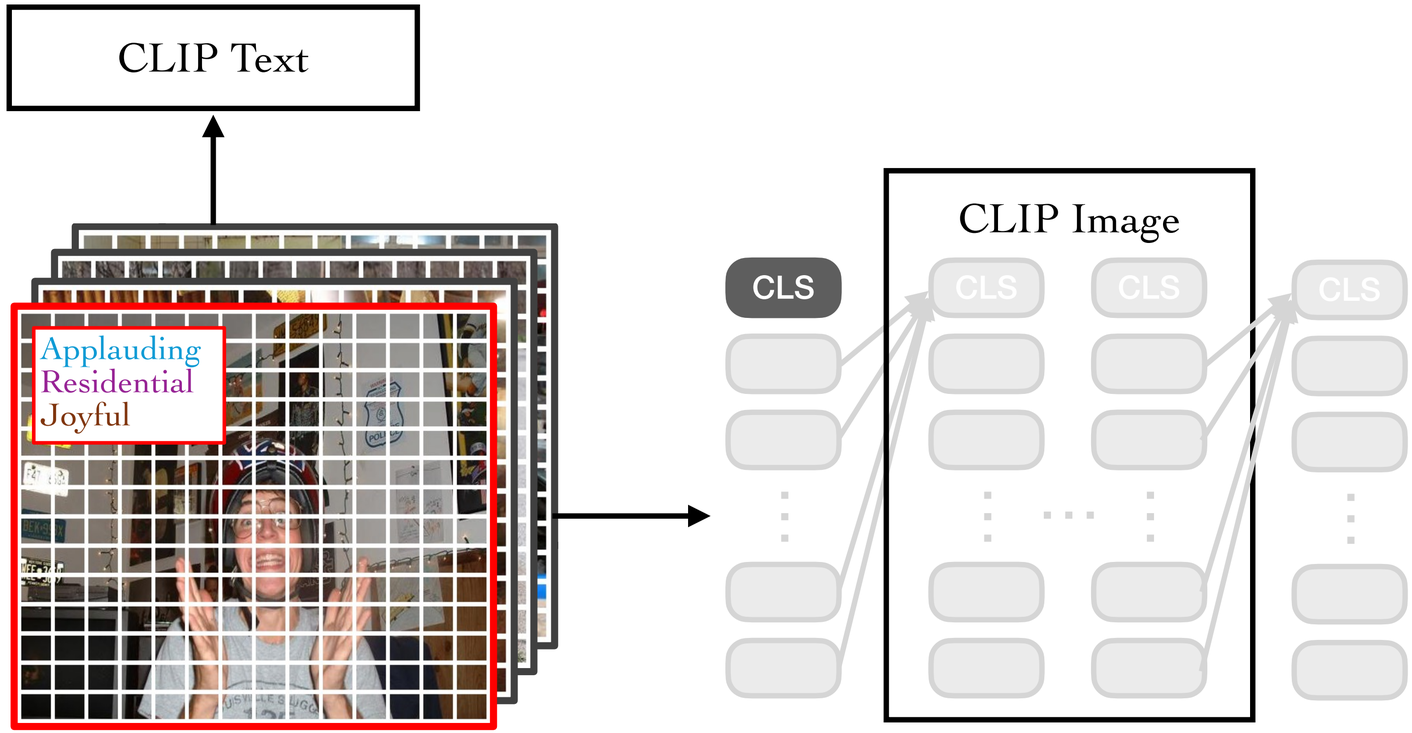
Dog accuracy: $\frac{90}{90+10} \approx 0.9$

Cat accuracy: $\frac{10}{80+10} \approx 0.1$

Overall accuracy: $\frac{90+10}{90+10+80+10} \approx 0.5$

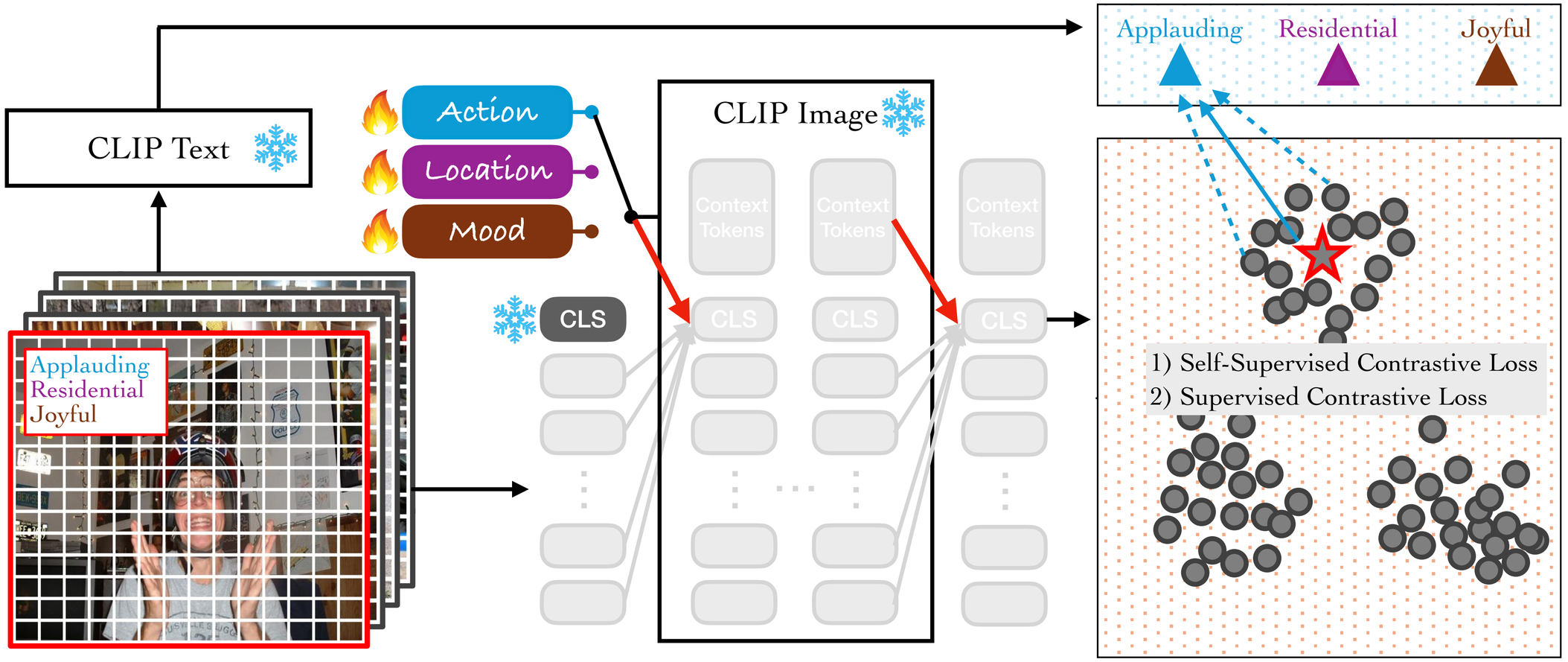
Our OAK Framework

1. Build on CLIP to leverage its general knowledge
2. Global context tokens to modulate the ViT for contextualized features



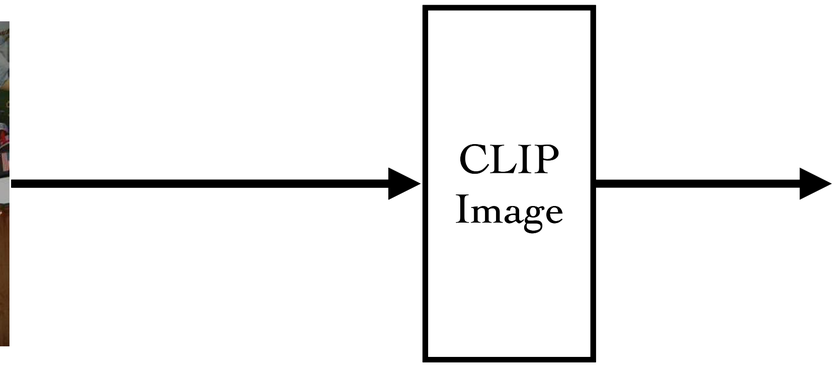
Our OAK Framework

1. Build on CLIP to leverage its general knowledge
2. Global context tokens to modulate the ViT for contextualized features
3. Bottom-up clustering of visual cues
4. Top-down guidance from semantic cues



Global Context Token → Individual Image Features

Traditionally



“person”

Global Context Token → Individual Image Features

Traditionally

CLIP Image

"person"

Action

Location

Mood

Ours

CLIP Image

"clapping"

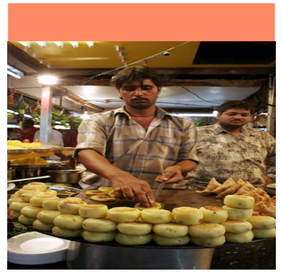
"residential"

"joyful"

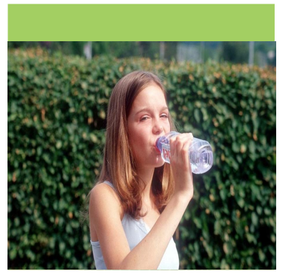
Evaluation Metrics

Unlabeled known

Unlabeled novel

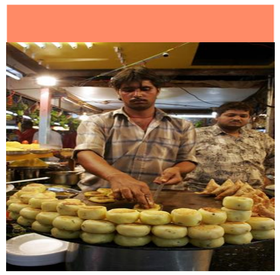


Labeled



Evaluation Metrics

Unlabeled known | Unlabeled novel



Action Groundtruth:	pushing a cart	drinking	cooking	reading
Action Predictions:	driving	drinking	driving	sleeping

Known accuracy = 1/2 = 0.5 | Novel accuracy = 0/2 = 0 | Overall accuracy = 1/4 = 0.25

Location Groundtruth:	urban	natural	store	residential
Location Predictions:	urban	natural	store	residential

Known accuracy = 2/2 = 1.0 | Novel accuracy = 2/2 = 1.0 | Overall accuracy = 4/4 = 1.0

Omni Accuracy:
How much images are correct in all contexts?

Omni known = 1/2 = 0.5 | Omni novel = 0/2 = 0 | Omni overall = 1/4 = 0.25

Semantic & Visual Cues are Both Necessary

Correct predictions across all contexts

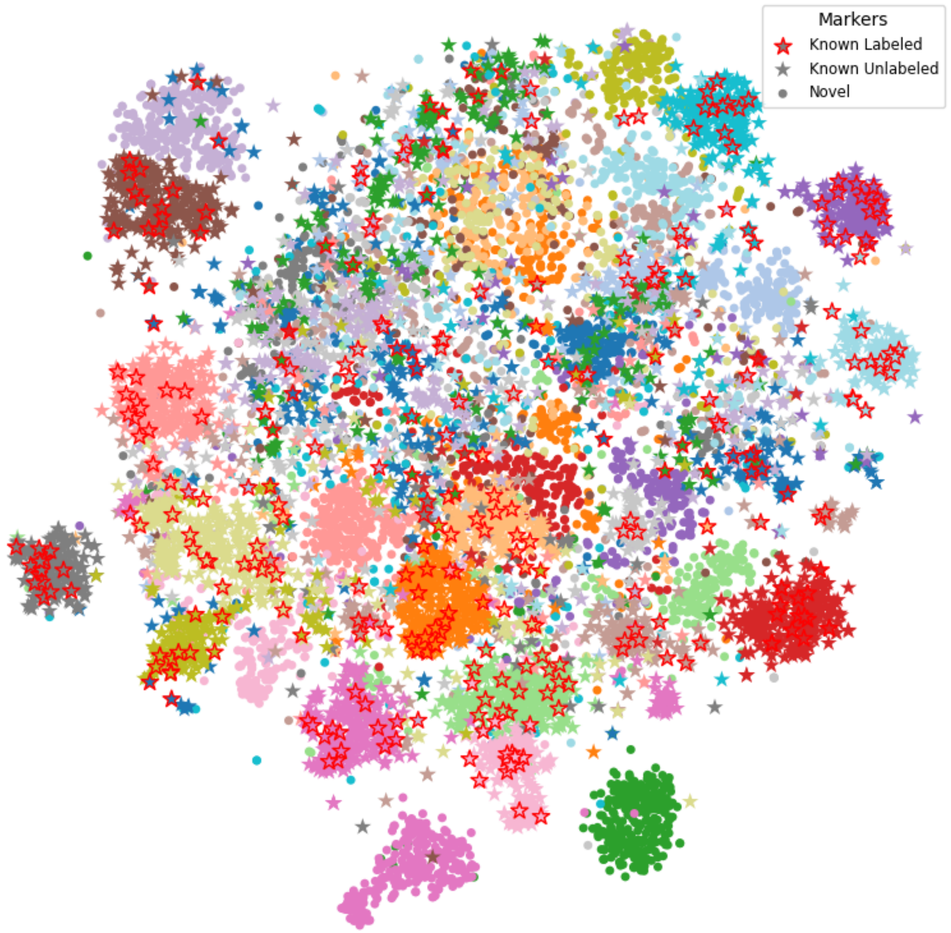
	Method	Known				Novel				Overall			
		Action	Location	Mood	Omni[a]	Action	Location	Mood	Omni	Action	Location	Mood	Omni
Semantic only	CLIP-ZS	**96.2**	60.4	**87.4**	75.0	-	-	-	-	-	-	-	-
	+ LLM vocab	93.5	58.3	75.9	75.0	38.6	34.2	35.4	0.0	65.2	47.5	55.0	43.0
	+ GT vocab	93.6	59.6	78.8	75.0	80.3	59.7	65.8	29.4	86.7	59.7	72.1	38.3
Visual only	SS-KMeans	67.0	56.3	43.2	0.0	53.9	71.1	78.2	35.3	60.2	62.9	61.3	47.7
	GCD	89.4	75.4	64.3	75.0	67.8	80.8	40.6	0.0	78.3	77.8	52.1	52.3
Semantic + Visual	OAK (ours)	88.9	**83.9**	68.8	0.0	**85.1**	**88.4**	**87.4**	**47.1**	**86.9**	**85.9**	**78.4**	**70.3**

[a] With only 8 images overlapping across all contexts, text is toned down to light gray due to unreliable results.

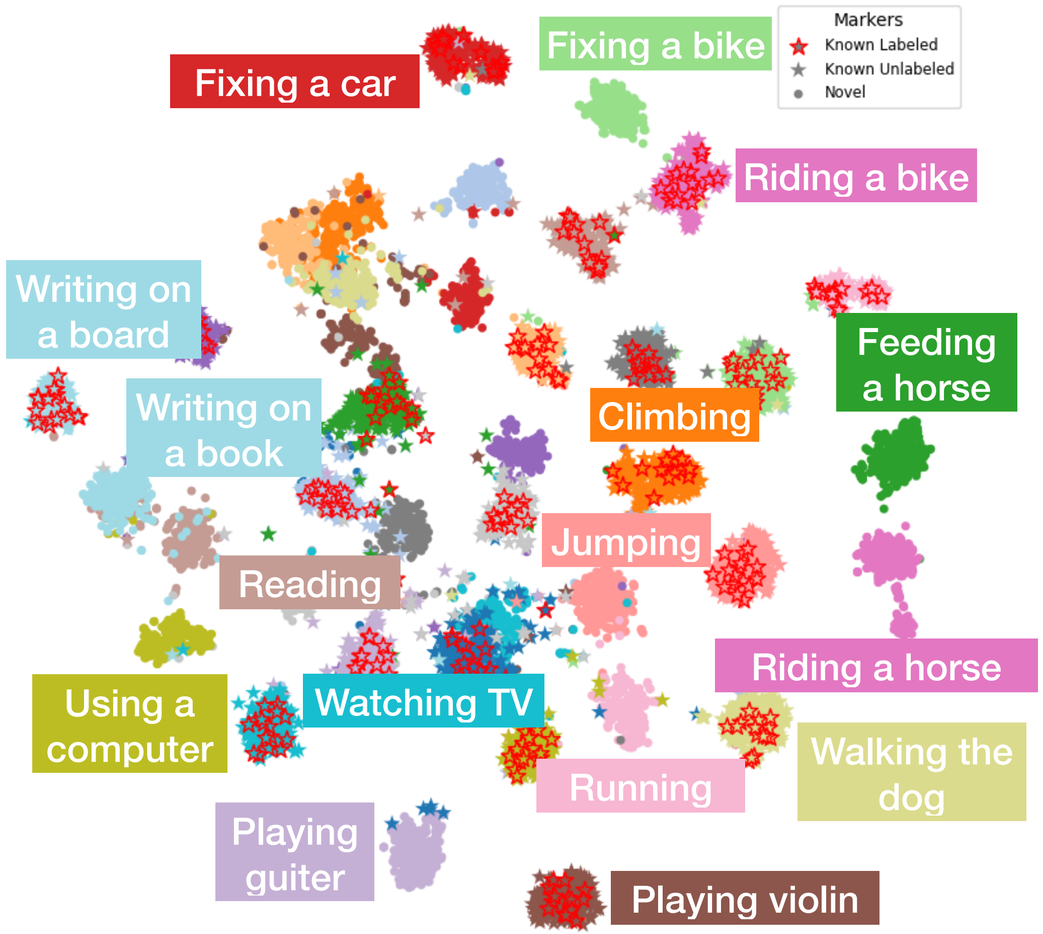
OAK Contextualizes Visual Features

Action

Before (CLIP)



After (OAK)



OAK Contextualizes Visual Features

Action

Before (CLIP)

playing guitar

playing guitar

playing violin

playing violin

playing guitar

After (OAK)

playing guitar

playing guitar

playing guitar

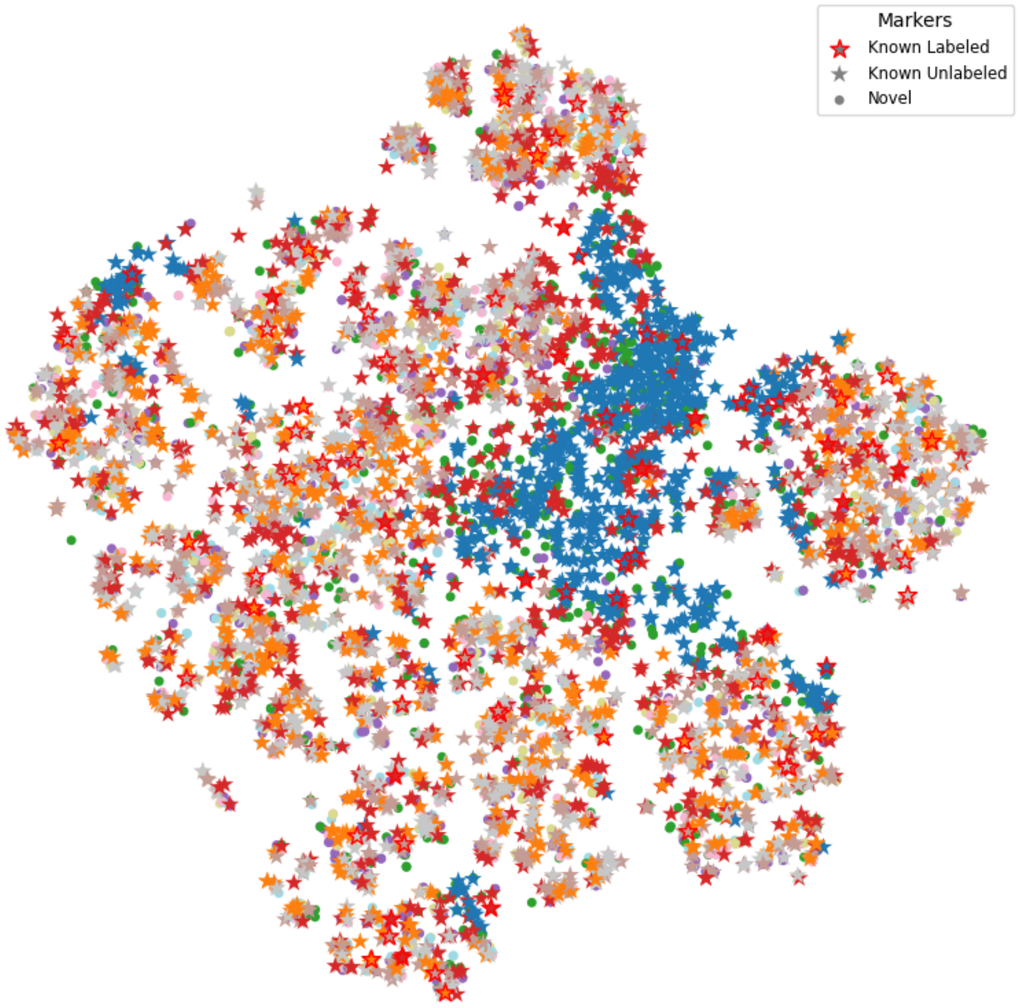
playing guitar

playing guitar

Novel Class

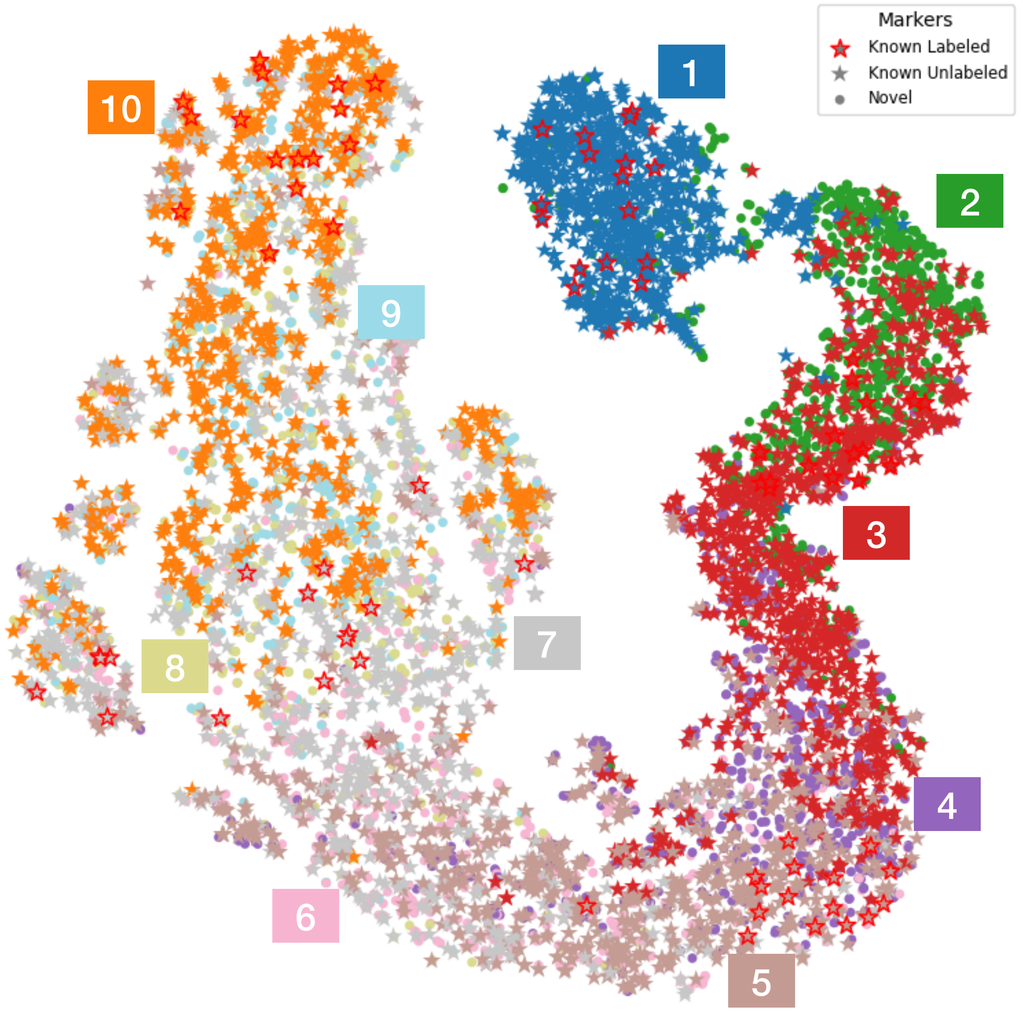
OAK Contextualizes Visual Features

Before (CLIP)



After (OAK)

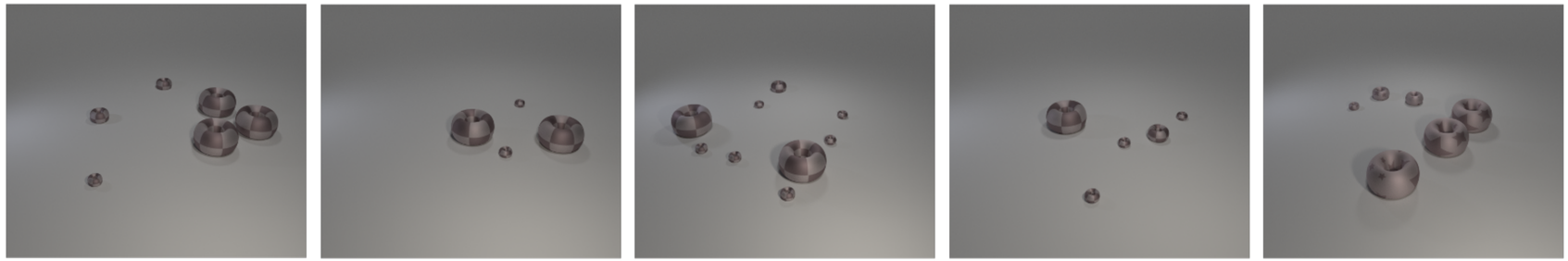
Count



OAK Contextualizes Visual Features

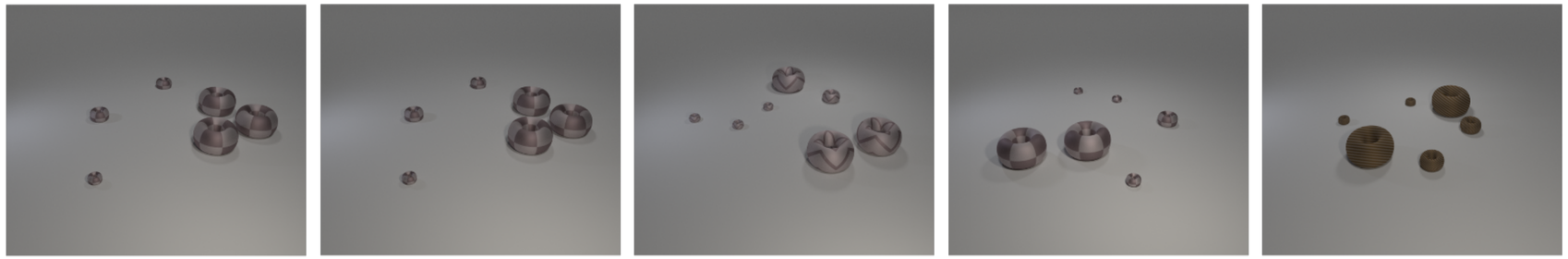
Count

Before (CLIP)



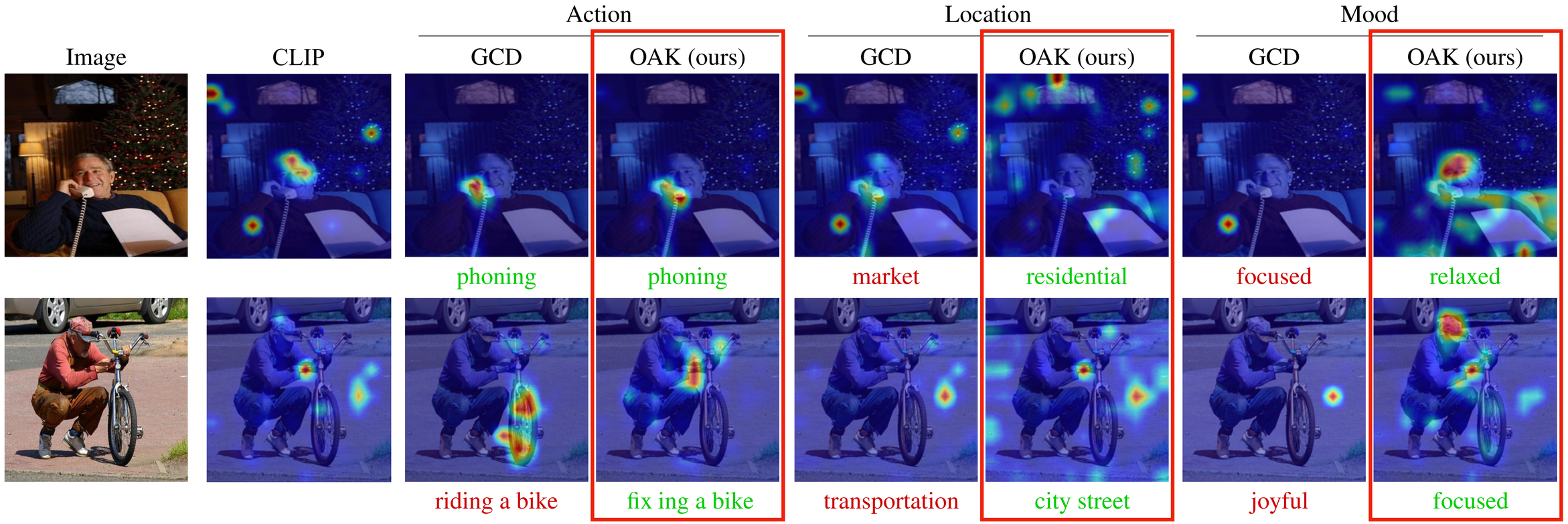
6 4 9 5 6

After (OAK)



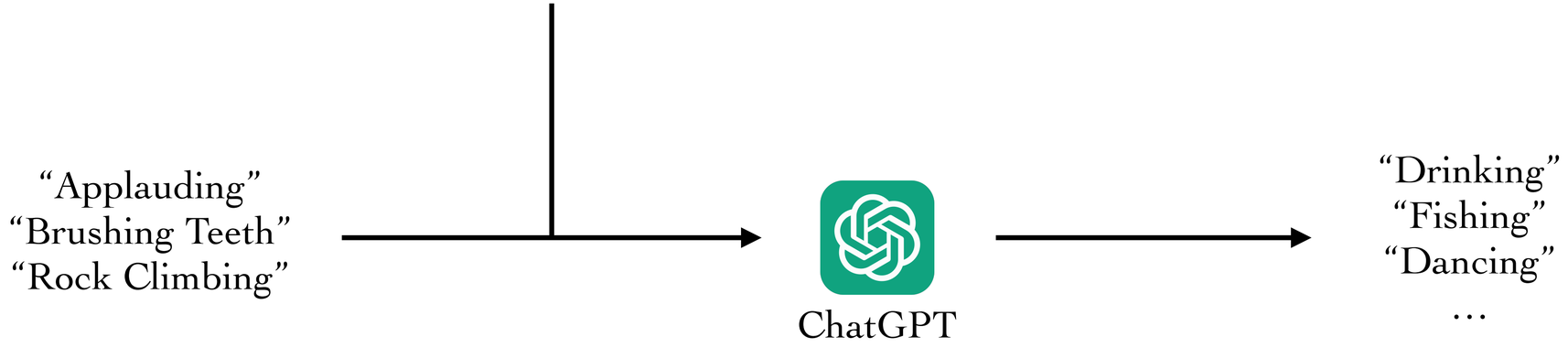
6 6 7 6 6

OAK Contextualizes Visual Attention

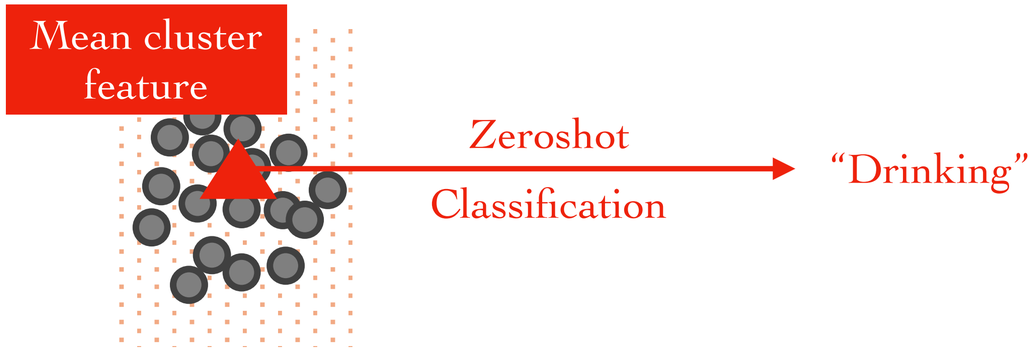


OAK Can Name the Discovered Clusters

I have a dataset of images from the following classes: ***[KNOWN_CLASSES]****. What are the most possible classes that will also be included in this dataset? Give me* ***[NUMBER_OF_NOVEL_CLASSES]*** *class names, only return class names separated by commas. Include quotation marks for each one.*



Constrained Text Inversion



OAK Can Name the Discovered Clusters

a) Stanford	Action		Location		Mood	
GT Label	cooking	reading	restaurant or dining area	urban area or city street	focused	relaxed
Prediction	preparing a meal	reading a book	commercial kitchen	suburban street	explorative	admiring

jumping named as *dancing*

pouring liquid named as *carrying a box*

Summary

- We are inspired to study a novel problem setting called open ad-hoc categorization, which learns categorization rules under varying contexts to predict predefined categories and discover new ones

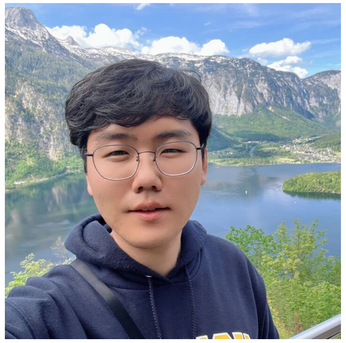
- We formulate the problem as predicting both known and novel concepts from unlabeled images using a few labeled examples within a given context.

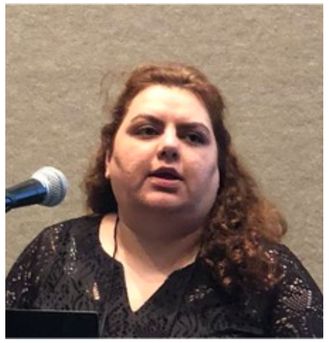
- We have two insights:

    1. Recognizing ad-hoc categories relies on the same perceptual processes as common categories;

    2. Novel concepts can be discovered semantically by expanding contextual semantics or visually by clustering similar patterns.

- A simple model that introduces a single learnable context token into CLIP, learned from combining CLIP's top-down open-vocabulary classification with GCD's bottom-up data-driven visual clustering





Zilin Wang*, Sangwoo Mo*, Stella X. Yu, Sima Bephour, and Liu Ren. Open Ad-hoc Categorization with Contextualized Feature Learning. In *CVPR*, 2025.

Contextualized Feature Learning

Action

"clapping"
"drinking"
"pushing a cart"
…

Location

"residential"
"urban"
"commercial"
…

Mood

"adventurous"
"joyful"
"happy"
…

"clapping"

"residential"

"joyful"

Application to Hierarchical Classification

Coarse Level

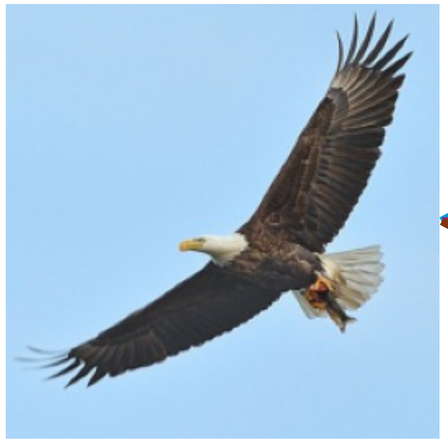
"bird"
"dog"
"insect"
…

Basic Level

"bird of prey"
"terrier dog"
"beetle"
…

Fine-Grained

"bald eagle"
"silky terrier"
"tiger beetle"
…



"bird"

"bird of prey"

"bald eagle"

Application to Embodied AI

Companion Robot

"talk"
"listen"
"remind"
…

Cleaning Robot

"wipe"
"sweep"
"dry"
…

Medical Robot

"monitor"
"alert"
"inject"
…

"listen"

"wipe"

"monitor"

Questions?